Instructions for Running the Code

Install the following Python libraries:

pip install Tensorflow, Numpy, Matplotlib, Scikit-learn

Dataset Setup

1. Download the Traffic Sign Dataset Classification dataset from Kaggle.
2. Add the dataset to your Kaggle Notebook.
3. Verify the dataset path:

/kaggle/input/datasets/ahemateja19bec1025/traffic-sign-dataset-classification/traffic_Data/DATA

Steps to Run the project

1. Open the Kaggle Notebook.
2. Add the Traffic Sign Dataset Classification dataset.
3. Run all notebook cells from top to bottom.
4. The program will:

- Load the dataset
- Preprocess images
- Train the CNN model
- Evaluate the model
- Generate accuracy and loss graphs
- Generate a confusion matrix
- Generate a classification report

5. After training is completed, the model will be saved as:

   traffic_sign_model.keras

Requirements

- Python
- TensorFlow
- NumPy
- Matplotlib
- Scikit-learn

Output

The project produces:

- Trained CNN Model

- Accuracy Graph
- Loss Graph
- Confusion Matrix
- Classification Report